

Python Full Stack (FastAPI) – Teaching Syllabus

Module 1: Introduction to Full Stack

- What is Full Stack Development?
 - Frontend vs Backend vs Database
 - Client-Server Architecture
 - HTTP Protocol
 - Request & Response
 - HTTP Methods (GET, POST, PUT, DELETE)
 - Status Codes
 - JSON
 - Installing FastAPI & Uvicorn
 - Creating the First FastAPI Application
-

Module 2: FastAPI Basics

- Project Structure
 - Routing
 - Path Parameters
 - Query Parameters
 - Request Body
 - Response
 - Response Models
 - API Documentation (Swagger UI)
-

Module 3: Frontend Integration

- Jinja2 Templates
- Static Files (HTML, CSS, JavaScript)
- Template Rendering
- Passing Data to Templates

- Navigation Between Pages
-

Module 4: Forms

- HTML Forms
 - GET & POST Forms
 - Receiving Form Data
 - Form Validation
 - Redirecting Users
-

Module 5: Database (MySQL)

- Connecting FastAPI with MySQL
 - Database Configuration
 - CRUD Operations
 - Create
 - Read
 - Update
 - Delete
 - SQL Queries
 - Error Handling
-

Module 6: REST APIs

- REST API Concepts
- API Endpoints
- Request Body
- Response Models
- Pydantic Models
- Testing APIs using Swagger UI
- Testing APIs using Postman

Module 7: Authentication

- User Registration
- User Login
- Password Hashing
- Session/JWT Authentication (Basic)
- Protected Routes
- Logout

Module 8: File Handling

- File Upload
- Image Upload
- Displaying Uploaded Files

Module 9: Git & GitHub

- Git Basics
- Repository Creation
- Clone
- Commit
- Push
- Pull

Module 10: Deployment

- Environment Variables
- Deploying FastAPI Application (Basic)

Suggested Projects

- Student Management System
- Library Management System

